

How do babies break into
the complexities of language?



T/Th 9:35-10:25am

Professor Casey Lew-Williams

Kennedy Casey, Abby Fergus, Sarah Joo, Chantal Valdivia, Claire Whiting

Midterm Exam

Median: 85

Highest score: 103

If you scored below 70, you can do an extra assignment to get your grade up to 70.

Today

Theories of language 🥊

Language is messy and fascinating 🔊

Infant-directed speech 👶

Learning about sounds 🗣️

[Word segmentation]

Theories of language

- **Universal Grammar (Chomsky, Pinker...)**

- Language cannot be learned through reinforcement/punishment.
- The general structures of the world's languages are similar because we're born with them; they're in the mind.
- We can produce sentences we've never heard before because we're born with some knowledge of grammar.

Colorless green ideas sleep furiously.

Green sleep colorless furiously ideas.



- **Learning theories (Bates, Elman, Clark, Goldberg, Saffran...)**

- The complexities of language are learnable via various domain-general and computationally feasible processes.
- The general structures of the world's languages are similar because learnable things stuck around and less learnable things went away over time.
- So, maybe infant learning mechanisms have shaped natural languages.



What language is...

according to Chomsky

The dog [that the cat scared] ran away.

The dog [that the cat [that the mouse [that the bug startled] surprised] scared] ran away.

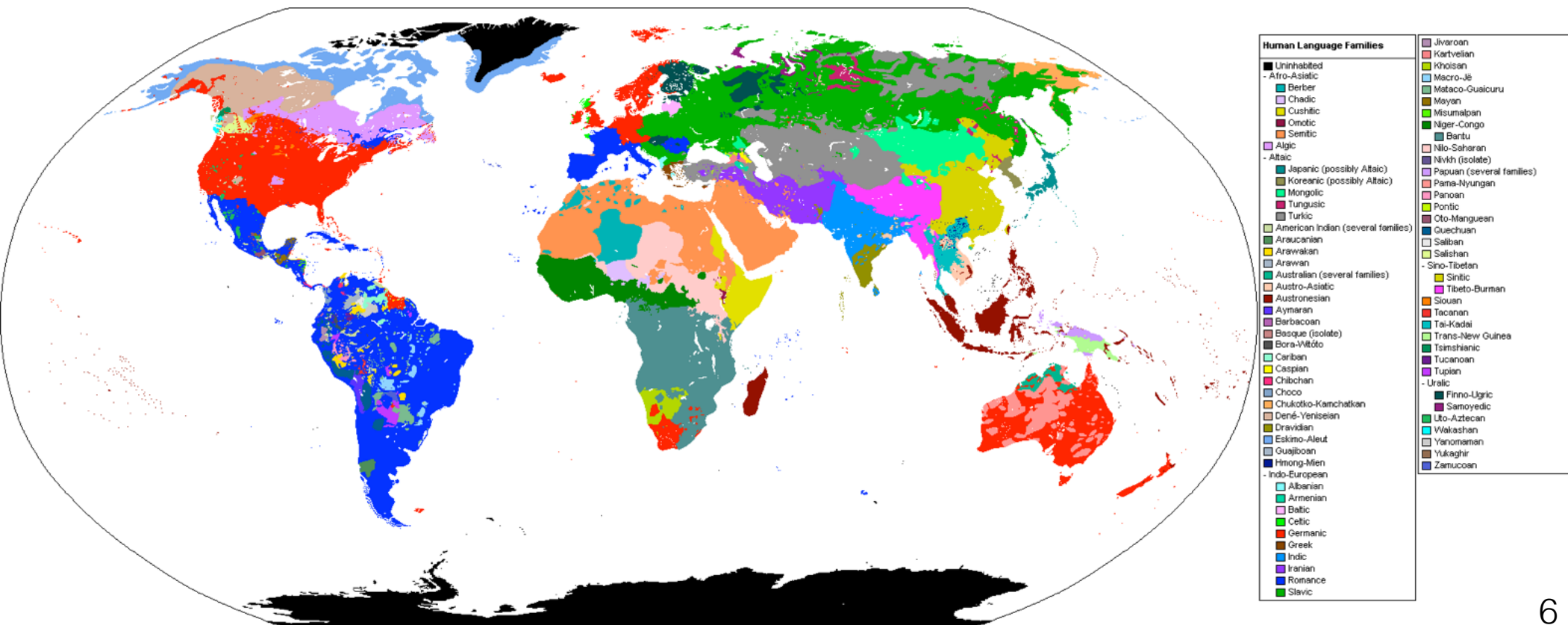
according to others (e.g., Bates, Elman, Clark, Goldberg, Saffran...)

A means of being understood, like playing charades. Full of restarts, messiness, and enormous variability.

**THESE ARE VERY DIFFERENT
WAYS OF THINKING ABOUT LANGUAGE**

Language: the most important tool ever invented.

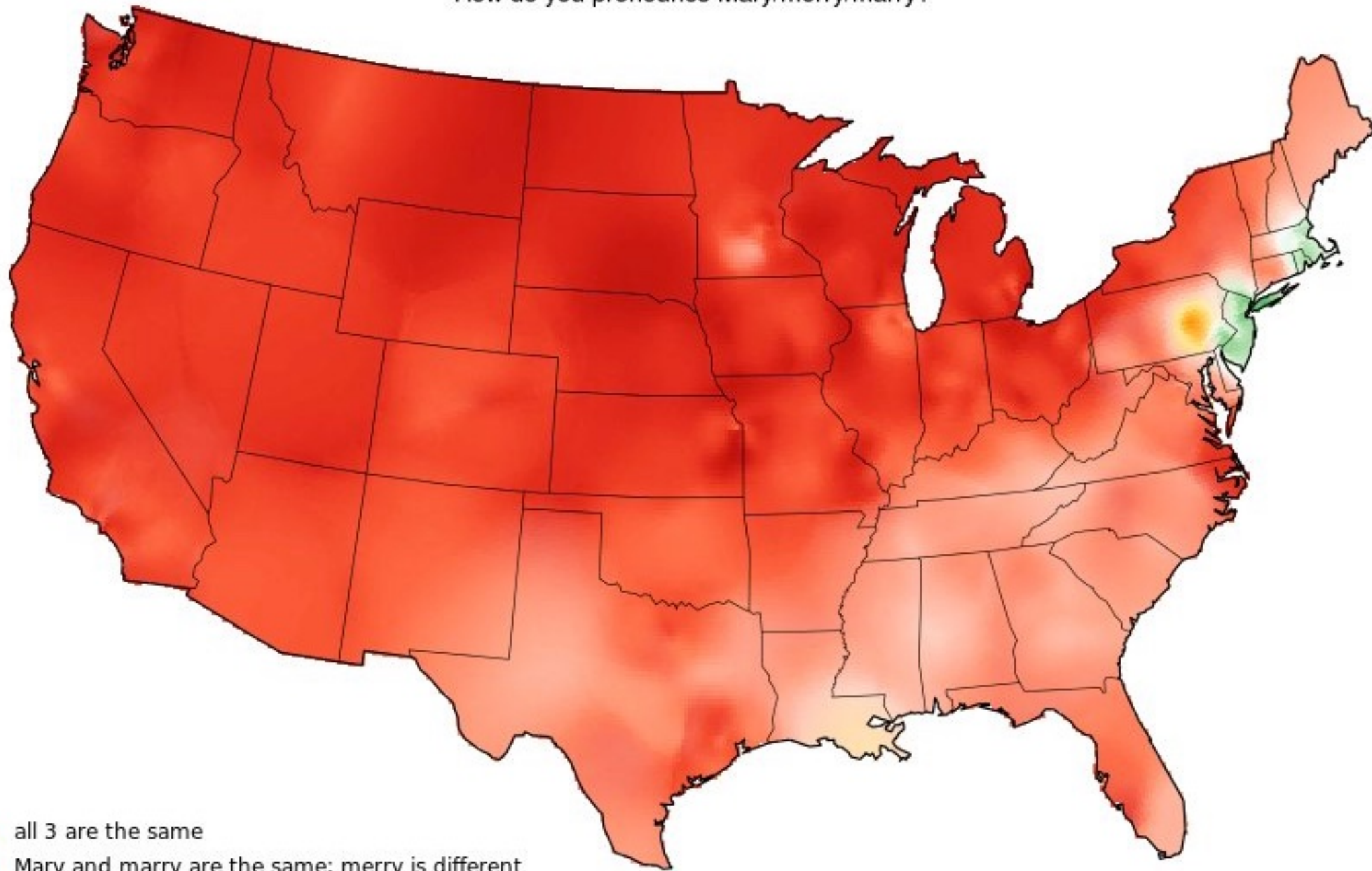
Provides us with a medium to think effectively, communicate with others, and collaborate with others.



One-slide crash course on the branches of linguistics

- **Phonetics/phonology**: the study of sounds
- **Morphology**: the study of smaller units within words
- **Semantics**: the study of language meaning
- **Syntax**: the study of rules that govern sentences
- **Pragmatics**: the study of meaning in context, of how language used to communicate

How do you pronounce Mary/merry/marry?



- all 3 are the same
- Mary and marry are the same; merry is different
- all 3 are different
- Mary and merry are the same; marry is different

Joshua Katz, Department of Statistics, NC State University

Ambiguous headlines

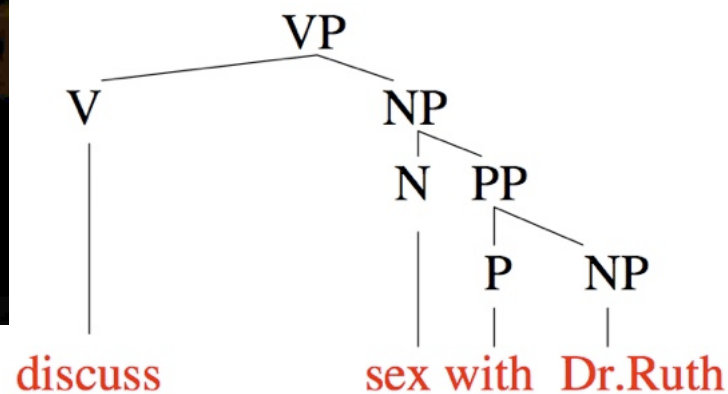
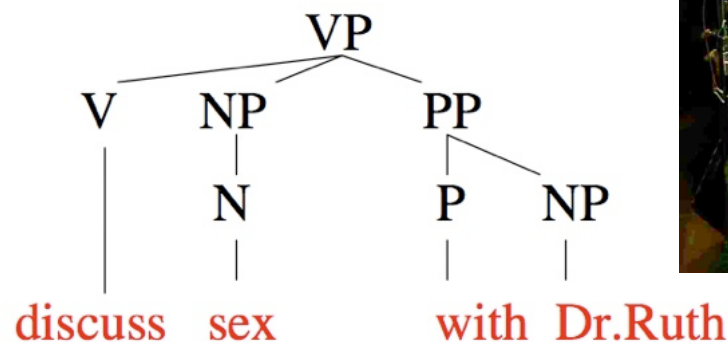


On Thursday, September 9, Gorman School hosted the first annual Grandparent's Day.

All Grandparents were invited to a school wide pancake breakfast. Upper grade students served as excellent chefs, as well as taking responsibility for serving the food and the clean up after-

Ambiguous headlines

- Michael headed for the court.
- Prostitutes appeal to Pope.
- 2 sisters reunited after 18 years at checkout counter.
- Grandmother of eight makes hole in one.
- On tonight's show, Letterman will discuss sex with Dr. Ruth.



If you have any information please call Crime Stoppers on 1800 333 000.

Correction

THERE was an error printed in a story titled "Pigs float down the Dawson" on Page 11 of yesterday's *Bully*.

The story, by reporter Daniel Burdon, said "more than 30,000 pigs were floating down the Dawson River".

What Baralaba pig-gery owner Sid Everingham actually said was "30 sows and pigs", not "30,000 pigs".

The Morning Bulletin would like to apologise for this error, which was also reprinted in today's *Rural Weekly CQ* before the mistake was known.

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30,000 pigs

30 sows and pigs

Lots of eggies



Each new cycle of tweens/teens makes language beautifully confusing



Sample of words/meanings that I've seen come into existence:

tubular, rad, like, hella, low-key (as an adverb), extra, mid, boujee, drip, skibidi...

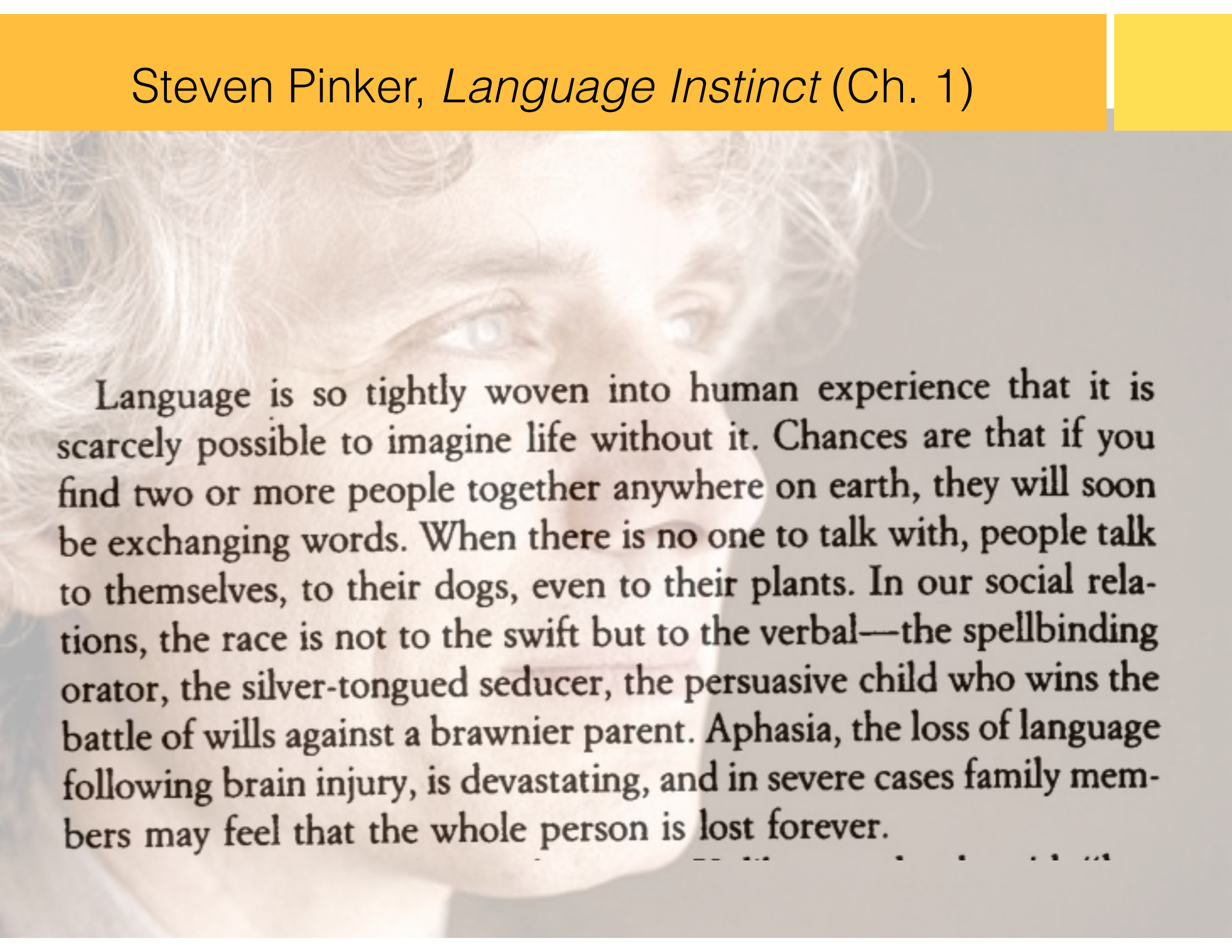
Toni Morrison on language

Oppressive language does more than represent violence; it is violence; does more than represent the limits of knowledge; it limits knowledge.



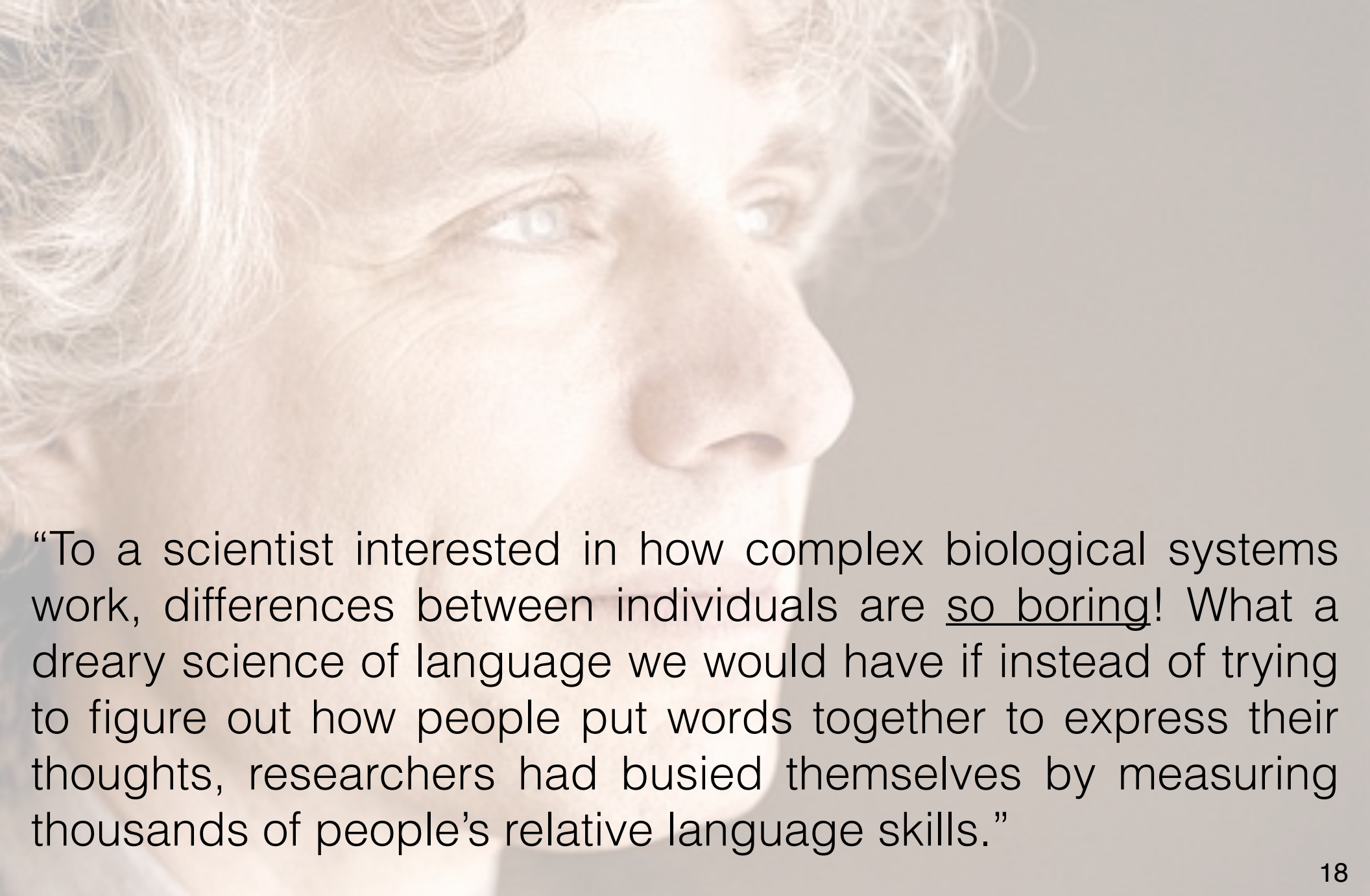
<https://www.nobelprize.org/prizes/literature/1993/morrison/lecture/>

Steven Pinker, *Language Instinct* (Ch. 1)



Language is so tightly woven into human experience that it is scarcely possible to imagine life without it. Chances are that if you find two or more people together anywhere on earth, they will soon be exchanging words. When there is no one to talk with, people talk to themselves, to their dogs, even to their plants. In our social relations, the race is not to the swift but to the verbal—the spellbinding orator, the silver-tongued seducer, the persuasive child who wins the battle of wills against a brawnier parent. Aphasia, the loss of language following brain injury, is devastating, and in severe cases family members may feel that the whole person is lost forever.

Steven Pinker, *Language Instinct* (Ch. 1)



“To a scientist interested in how complex biological systems work, differences between individuals are so boring! What a dreary science of language we would have if instead of trying to figure out how people put words together to express their thoughts, researchers had busied themselves by measuring thousands of people’s relative language skills.”





Language Learning



Stuff kids say

E.B. White's child:

“What did you bring that book that I don't like to be read to out of up for?”

Stuff kids say

age 2: "This is my jigsaw. I'm sawing some jig."



age 3: (pointing to the back of her knee): "My legpit hurts."

Stuff kids say

Kid at 3.5 years:

Is everybody as tired as I am? I'm 150 tired...I'm 199 tired.

You're very lucky. You're 20 hundred and 750 lucky.

Stuff kids say

At 19 months old, my daughter could only say 4 words. We were worried.

Fast forward four short months: on the way out of the pediatrician's office, she turned to me and said, "You know, Dad, what I like about going to the doctor's office is getting to play with all of the toys in the waiting room."

Stuff kids say

My 7-year-old Spanish student, who was bilingual in French and English and was learning Spanish, once said:

Esa voiture roja pasó cerca (meaning ‘that red car passed close (to us)’)

(She didn't remember the word in Spanish for car. Instead of saying *carro/coche*, she used a modified version of the French word *voiture*, which is feminine, and she used the feminine adjectives for agreement perfectly.)

"My mind is very angry,
and so am I"

Stuff kids say

When my son was 3, he couldn't remember the word 'vet/veterinarian', so he said "Let's pretend we're cat alivers"

Stuff kids say

After she made a few not-so-nice comments, I told my 8-year-old that I didn't want to hear her say anything negative.

She said "1 minus 5."

Stuff kids say

Mom realizes her missing sunglasses are on her head:
“I’m not always the brightest crayon in the box”

45 minutes later...

Two-year-old is struggling to put a train track together:
“I’m not always the goodest crayon at making trains.”

Stuff kids say

My own son, at 1 year 9 months:

“Pat-a-cake, pat-a-cake, baker's man,
Bake me a cake as fast as you can.
Roll it, pat it, and mark it with a B.
Put it in the oven at 350.”

Today

Theories of language 🥊

Language is messy and fascinating 🔊

Infant-directed speech 👶

Learning about sounds 🧠💡

[Word segmentation]

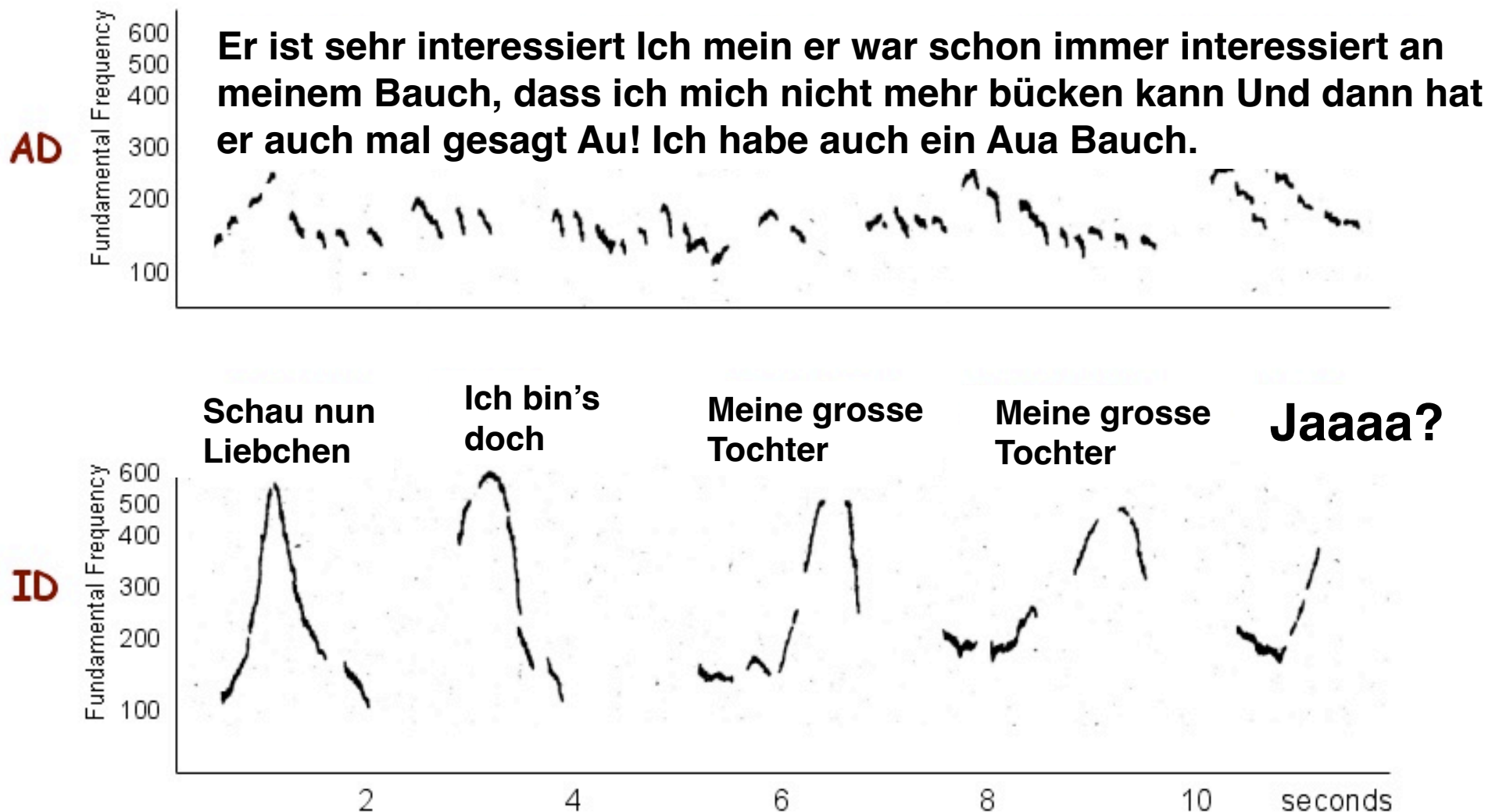
Language milestones

- Note that I'm not devoting much attention to specifics about language milestones. That's in the book.
- There's too much variability in language learning to pinpoint milestones in a meaningful way.

Infant-directed speech

Infant-directed speech

- The speaking style used by adults when talking to infants
- Highly variable pitch, shorter sentences, more pauses
- Exists in most languages, though with great variability

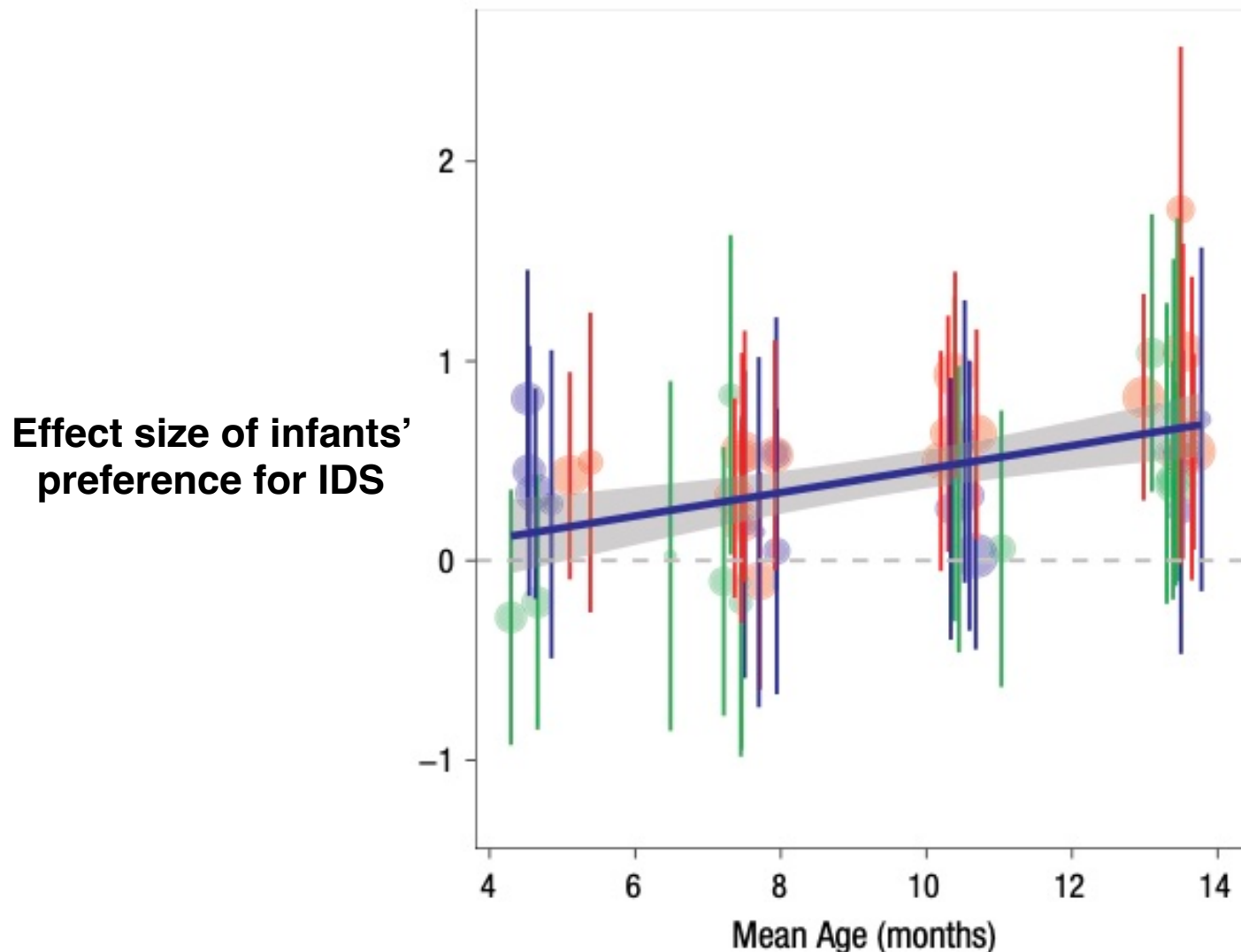




Infant-directed speech

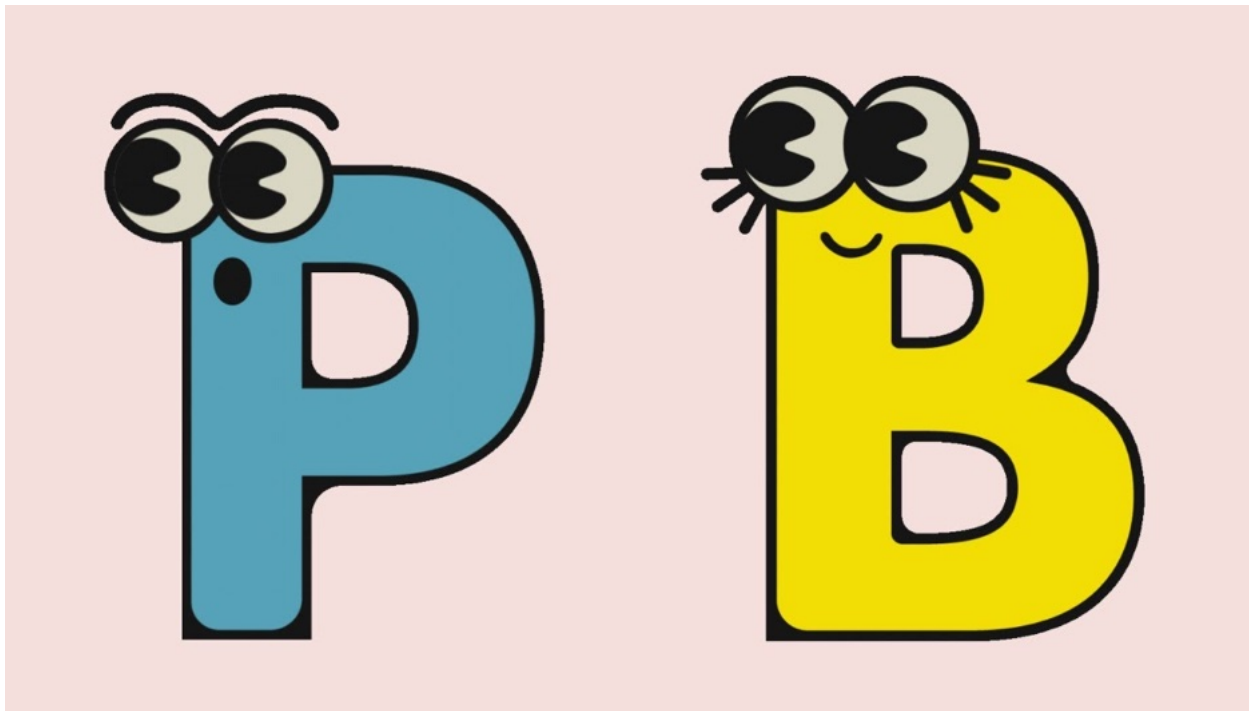
Why do we use infant-directed speech?

- It gets infants' attention. Infants prefer it over ADS.



Why do we use infant-directed speech?

- It helps infants distinguish phonemes and helps distinguish talkers, even by 1 month of age.



Why do we use infant-directed speech?

- It conveys emotion and meaning



Participants: 9- and 18-month-olds

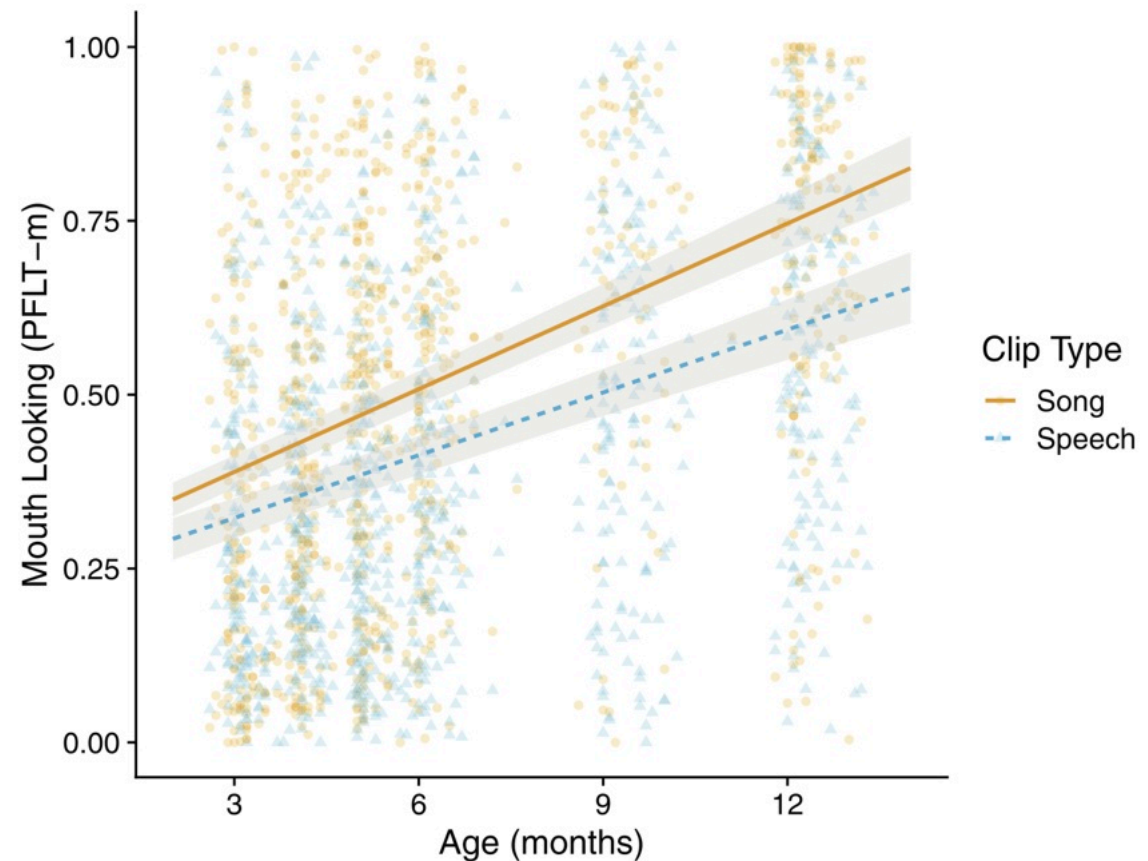
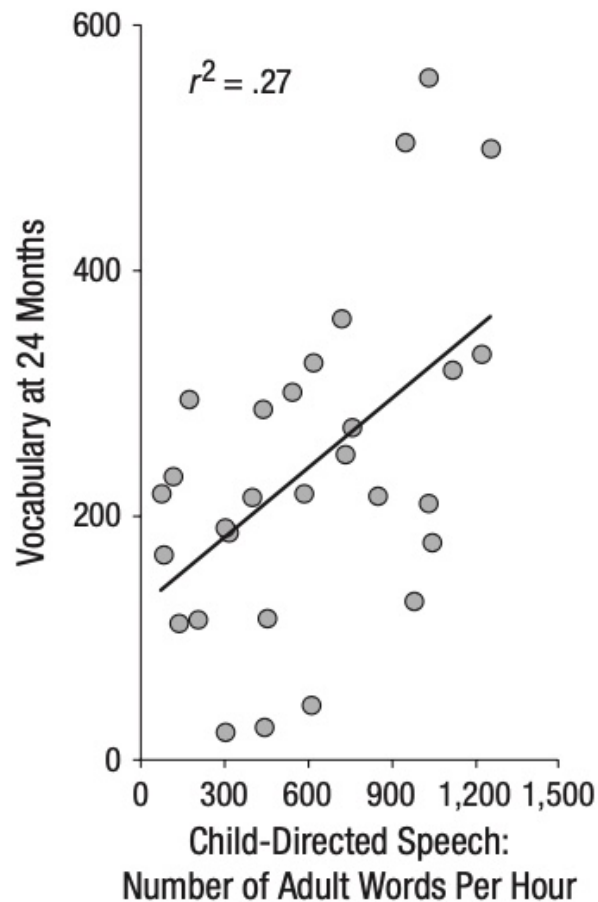
Procedure: played with novel toys and heard 4 types of sentences:

positive vs. negative words
soothing vs. harsh tone

INFANTS' DEVELOPING
SENSITIVITY TO
"MIXED MESSAGES"

Why do we use infant-directed speech?

- More IDS = larger vocabulary size and increased looking to the caregiver's mouth



Why do we use infant-directed speech?

- Supports early imitation?

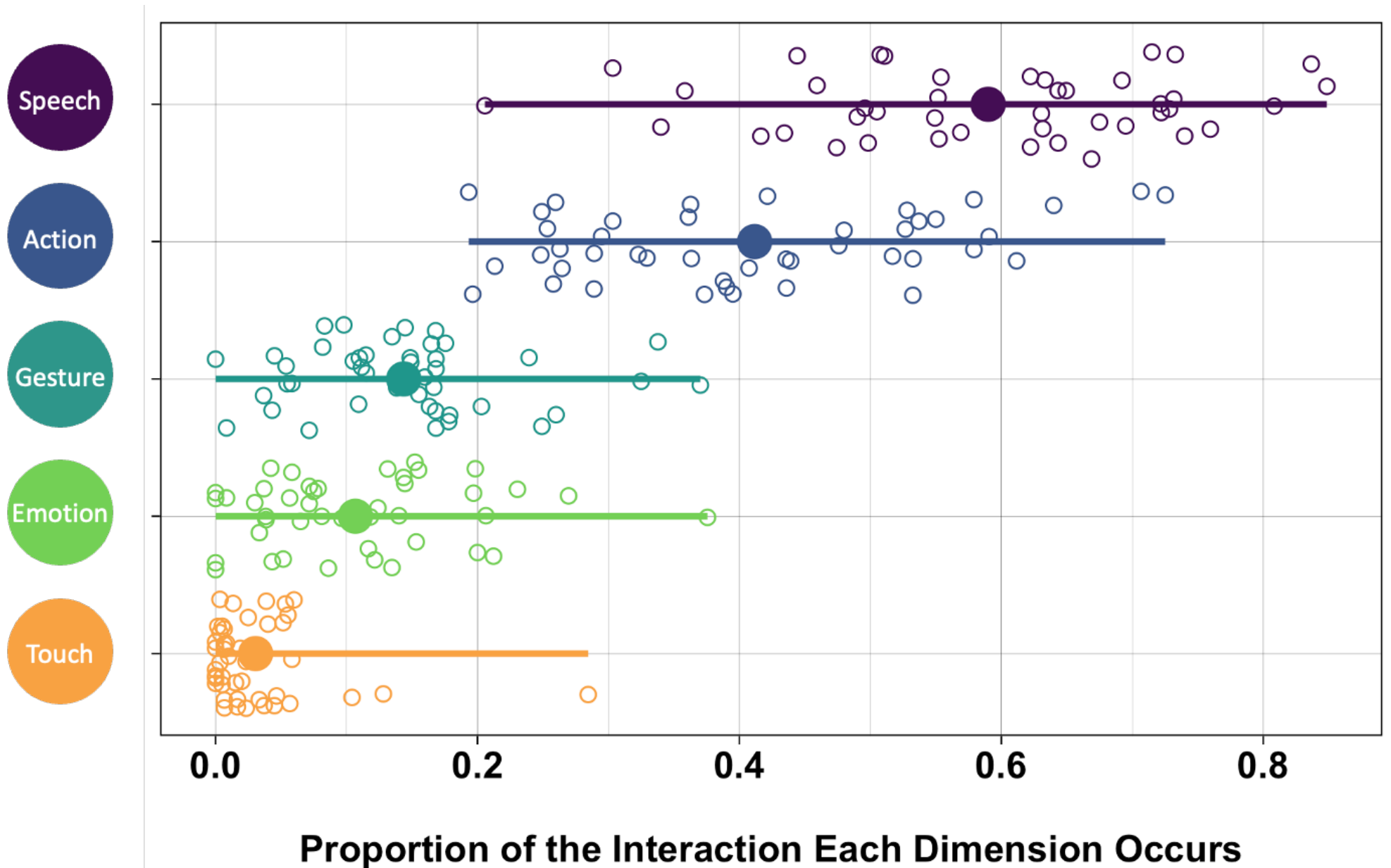


Why do we use infant-directed speech?

- Boosts brain development (mature white matter connectivity) in a key language area (left arcuate fasciculus)



Communication with infants is much more than speech alone



Other animals use infant-directed speech too



Zebra finch

(Chen et al., 2016)



Bats

(Fernandez & Knörnschild, 2020)



Great apes

(Knox et al., 2019)

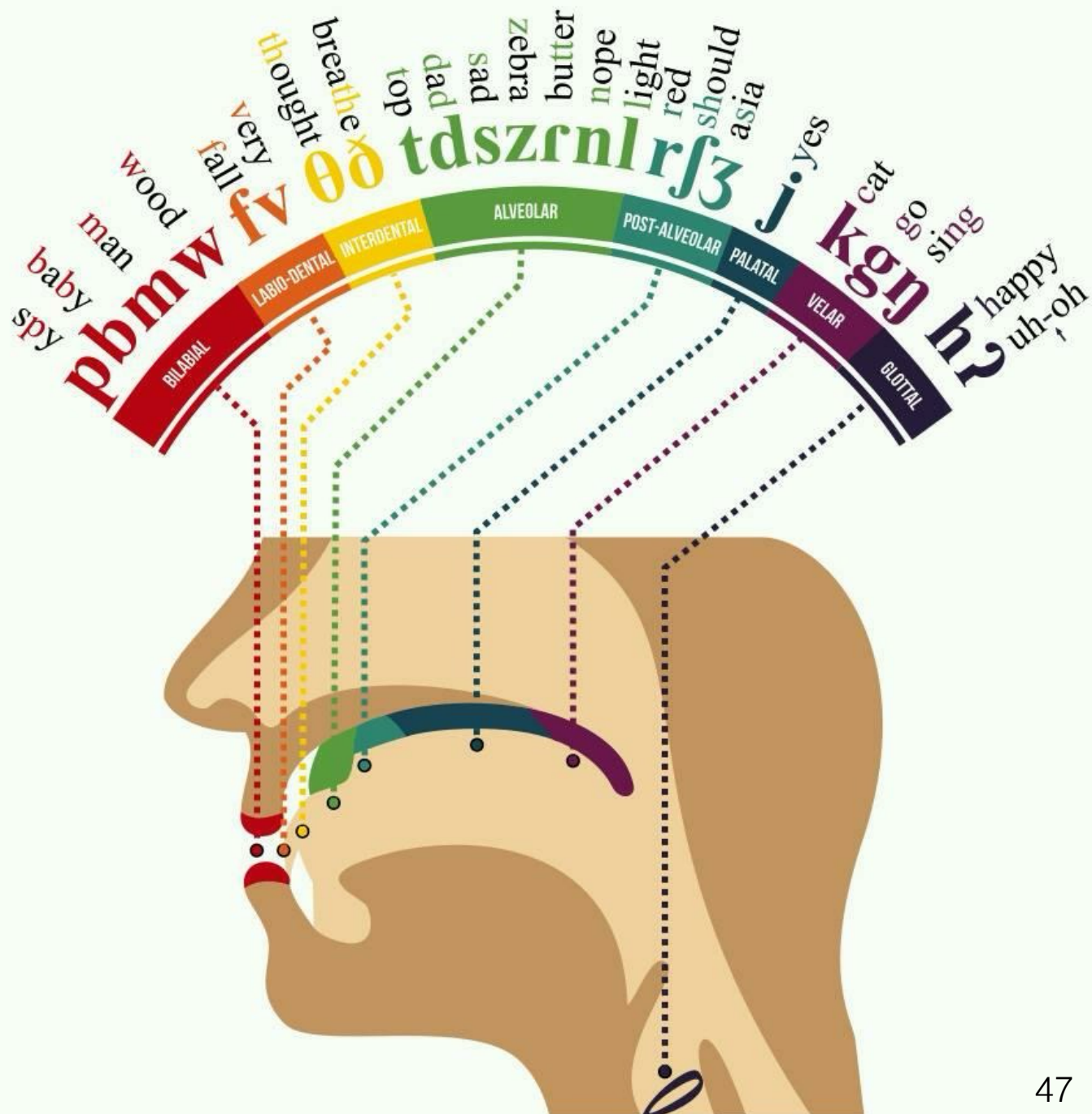


Dolphins

(Sayigh et al., 2023)

Learning about sounds

Sounds & the vocal tract



Bilabials: p, b...

Alveolars: t, d...

Velars: k, g...

Beatboxing during an MRI



Sounds matter



<https://youtube.com/shorts/7Pjl3KH029s?si=FOE9RL5b4v0OgXSh>

Learning sounds

The tones of Cantonese

To hear the sounds press [Play](#)

CANTONESE si			
Chinese Character	Tone symbol	Tone description	English gloss
詩	˥ 55	high level	'poem'
試	˨˨ 33	mid level	'to try'
事	˨˨ 22	low level	'matter'
時	˨˨˨ 21	low falling	'time'
使	˨˨˨˨ 24	low high	'to cause'
市	˨˨˨˨ 23	low mid	'city'

<http://www.phonetics.ucla.edu/vowels/chapter2/cantonese/recording2.2.html>

Learning sounds

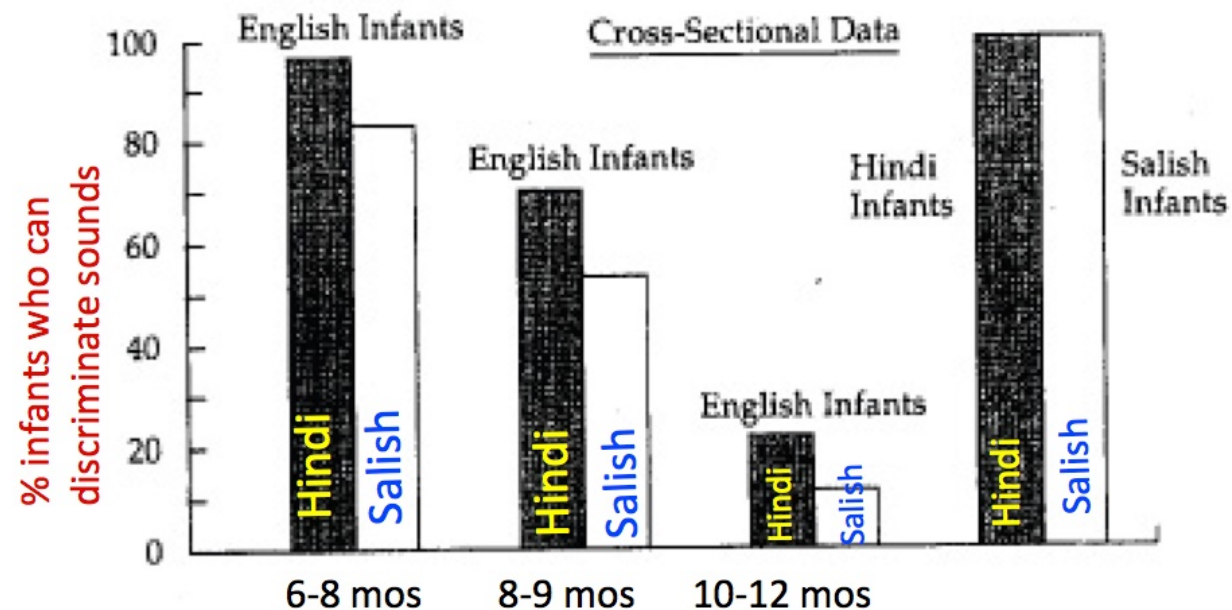
Tough
Trough
Though
Thought
Through
Thorough
Throughout

Learning sounds



Tuning perception to the native language

- Newborns distinguish phoneme categories (i.e., sounds that make meaningful distinctions, such as rake vs. lake), even for sounds they've never heard.
- Test languages: Nlaka'pamux (a.k.a. Salish) & Hindi
- Sounds: Salish /'ki/ and /'qi/, Hindi /ta/ sounds



Tuning perception to the native language

- This effect has been observed across many languages...

Hindi retroflex-dental (Anderson et al., 2003; Rivera-Gaxiola et al., 2002)

Salish ejectives and Zulu clicks (Best et al., 1994 and so many more)

Chinese fricatives; English laterals (Kuhl et al., 2002)

Spanish consonants (Rivera-Gaxiola et al., 2005)

English and Swedish vowels (Kuhl et al., 1992)

Estonian vs. Finnish vowels (Cheour et al., 1998)

Catalan vowels (Bosch & Sebastián-Gallés, 2003)

Lexical Tone (Mattock & Burnham, 2006; 2008; *Liu & Kager, 2014)

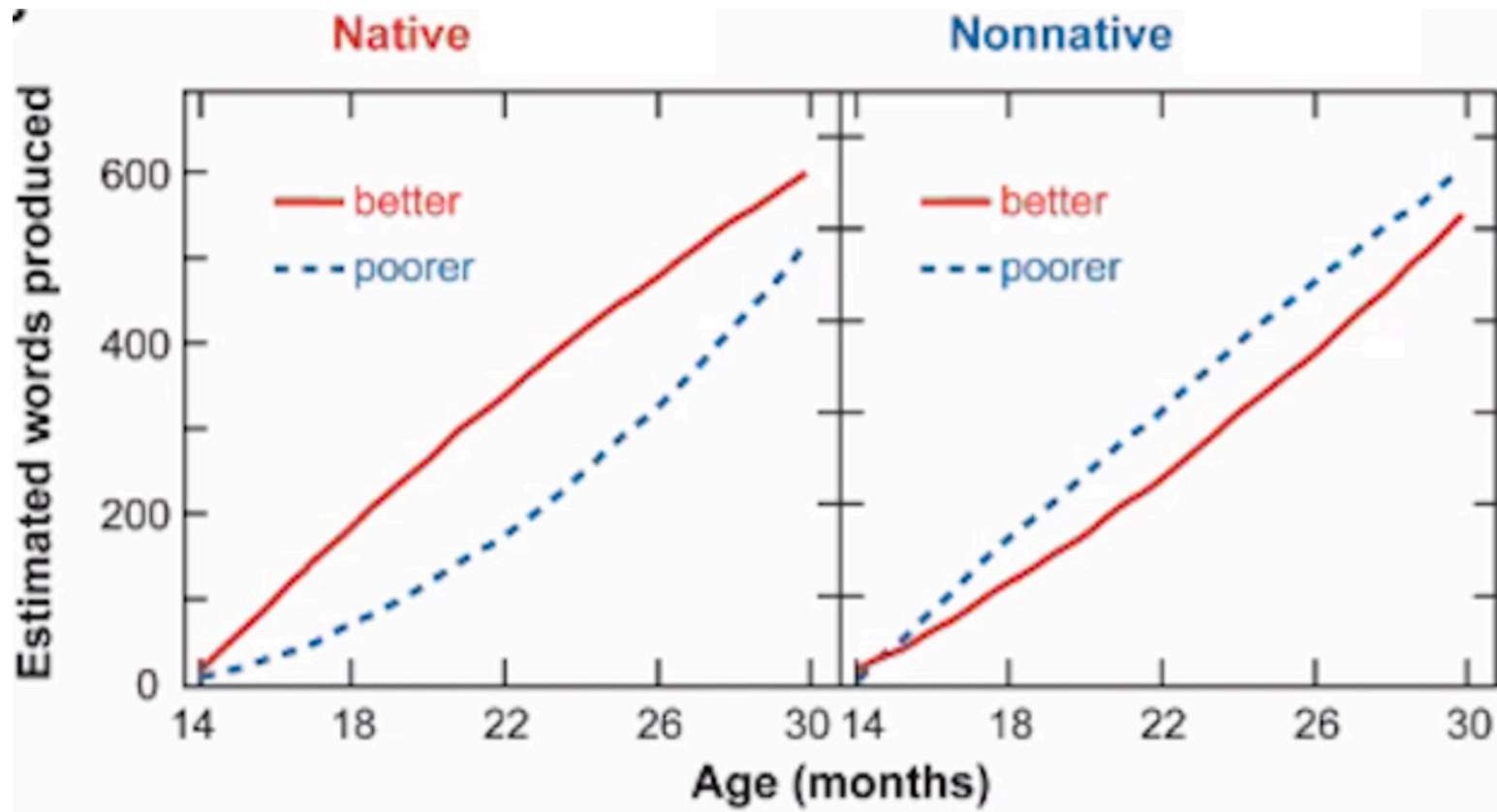
Sign Language (Baker et al., 2009; Palmer et al., 2012)

Auditory-Visual Speech Matching (Pons et al., 2012)

Bilingual infants maintain both (Bosch & Sebastian-Galles, 2003; Burns et al., 2007) AND discriminate dif across their languages (Sundara et al., 2008)

Tuning perception to the native language

- ERP study: If an infant is better at discriminating sounds in a non-native language, their vocabulary is lower. Why?



Perspective-taking exercise: Xhosa!



Word segmentation
(you'll discuss this in your precept)

Agglutinative languages present
a word segmentation challenge

Avrupalılaştıramadıklarımızdan mısınız?

Translation from Turkish:

“Are you one of those whom we could not Europeanize?”

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Torno a casa con le bici cariche di frutta in bilicos sulla sella. Per ascoltare la fuga quasi cadisul melo e inciampi sulla biro sull'erba. Il picchioso abituato a fare la sua casa in ogni melo cava e alto. Diraduna bici in rapida fuga in correlamento obliqua aerea. La cavia Bidae in fuga da casa per aver giocato con le bilie blu. La bici ha subito un danno dentro la casa del capodiLara. Se porti il melo sulla bici forse cala un po' di chili. Il tuo melo arcano fuga' a fache debilita la folla. Gusto di bigoli dentro casa o coricata all'ombra del melo verde. Arriviamo in bici fino al bivvi del grande melo con un caro amico. L'azia Carolasie' esibita in una fuga colla bici verde. La bici in lenta fuga dal giardino capitain casa mia. Torno a casa con le bici cariche di frutta in bilicos sulla sella. Per ascoltare la fuga quasi cadisul melo e inciampi sulla biro sull'erba. Il picchioso abituato a fare la sua casa in ogni melo cava e alto. Diraduna bici in rapida fuga in correlamento obliqua aerea. La cavia Bidae in fuga da casa per aver giocato con le bilie blu. La bici ha subito un danno dentro la casa del capodiLara. Se porti il melo sulla bici forse cala un po' di chili. Il tuo melo arcano fuga' a fache debilita la folla. Gusto di bigoli dentro casa o coricata all'ombra del melo verde. Arriviamo in bici fino al bivvi del grande melo con un caro amico. L'azia Carolasie' esibita in una fuga colla bici verde. La bici in lenta fuga dal giardino capitain casa mia.

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$$pr y|x = \frac{\text{freq } xy}{\text{freq } x}$$